Lecture 13: Proportions and Categorical data

BIOS 600 - Fall 2025

UNC - Chapel Hill

Announcements

- Lab 5 session today. Watch the video ahead of time and attend at 3:30 if you need help.
- Lab 5 due Tuesday 10/21 at 11:59pm (due to fall break)
- No HW until next week - enjoy your fall break!

Overview

- Review AE 03 from last time.
- Chi-square test and example
- Fisher exact test and example

Very optional reading

- P&G Chapters 14, 15
- OI: Sections 5.2, 5.3, 6.1, 6.2, 6.4

Application Exercise

AE 03

In small groups, take a few minutes to review AE 03.

- If you're already done, take turns presenting the exercises to a partner (explain the code, explain how you got your conclusion.)
- If you still need some time to work on it, take this time to do so.

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Pneumococcal disease

The World Health Organization estimates that in developing countries 814,000 children under the age of five die annually from invasive pneumococcal disease (IPD), with an estimated 1.6 million deaths affecting all ages globally.

Several recent studies have identified associations between pneumococcal serotypes (species variations) and patient outcomes from IPD. We consider data from a study of pneumococcal serotypes and mortality (Inverarity et al. (2011)).

Pneumococcal disease

Serotype	Alive	Dead	Total
Serotype 10	37	7	44
Serotype 31	24	10	34
Total	61	17	78

Is there an association between serotype and mortality?

Chi-square tests

H_0: There is no association between serotype and mortality

H_1: There is an association between serotype and mortality

Under H_0, we have independence between serotype and mortality, implying that joint probabilities should be equal to products of marginal distributions:

Serotype	Alive	Dead	Total
Serotype 10	?	?	44
Serotype 31	?	?	34
Total	61	17	78

What counts might we expect under H_0?

Intuition: Joint and Marginal Probabilities

- Suppose we look at two variables: **Serotype** and **Mortality** (Alive or Dead).
- The **joint probability** describes the probability of being in *both* categories (e.g., "Serotype 10 **and** Alive").
- The **marginal probabilities** describe totals across one variable:
  - P(Serotype 10) = proportion of all patients with Serotype 10
  - P(Alive) = proportion of all patients who survived

Marginal totals

Serotype	Total
10	44
31	34
Total	**78**

$P(\text{serotype } 10) = \frac{44}{78}$

Marginal totals

$P(\text{Alive}) = \frac{61}{78}$

Outcome	Total
Alive	61
Dead	17
Total	**78**

What would happen if these two variables were independent?

- Under the **null hypothesis of independence**, the chance of being both "Serotype 10" **and** "Alive" should just be:

$$P(\text{Serotype 10 and Alive}) = P(\text{Serotype 10}) \times P(\text{Alive})$$

- We can multiply these probabilities by the total number of observations to find the **expected counts**.

For example:

Expected Count (Sero. 10, Alive)

$$\text{"}E_{10,Alive} = P(\text{Serotype 10}) \times P(\text{Alive}) \times N$$

- Using our totals:

$$P(\text{Serotype 10}) = \frac{44}{78}, \quad P(\text{Alive}) = \frac{61}{78}, \quad N = 78$$

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Compute expected count under independence

```r
# Compute expected count under independence
expected_10_alive <- (44/78) * (61/78) * 78
expected_10_alive
```

```
[1] 34.41026
```

- We can repeat this for each cell to find what we'd expect if serotype and mortality were truly unrelated.
- The chi-square test then measures how far our observed counts deviate from these expected counts.

O = observed
E = expected

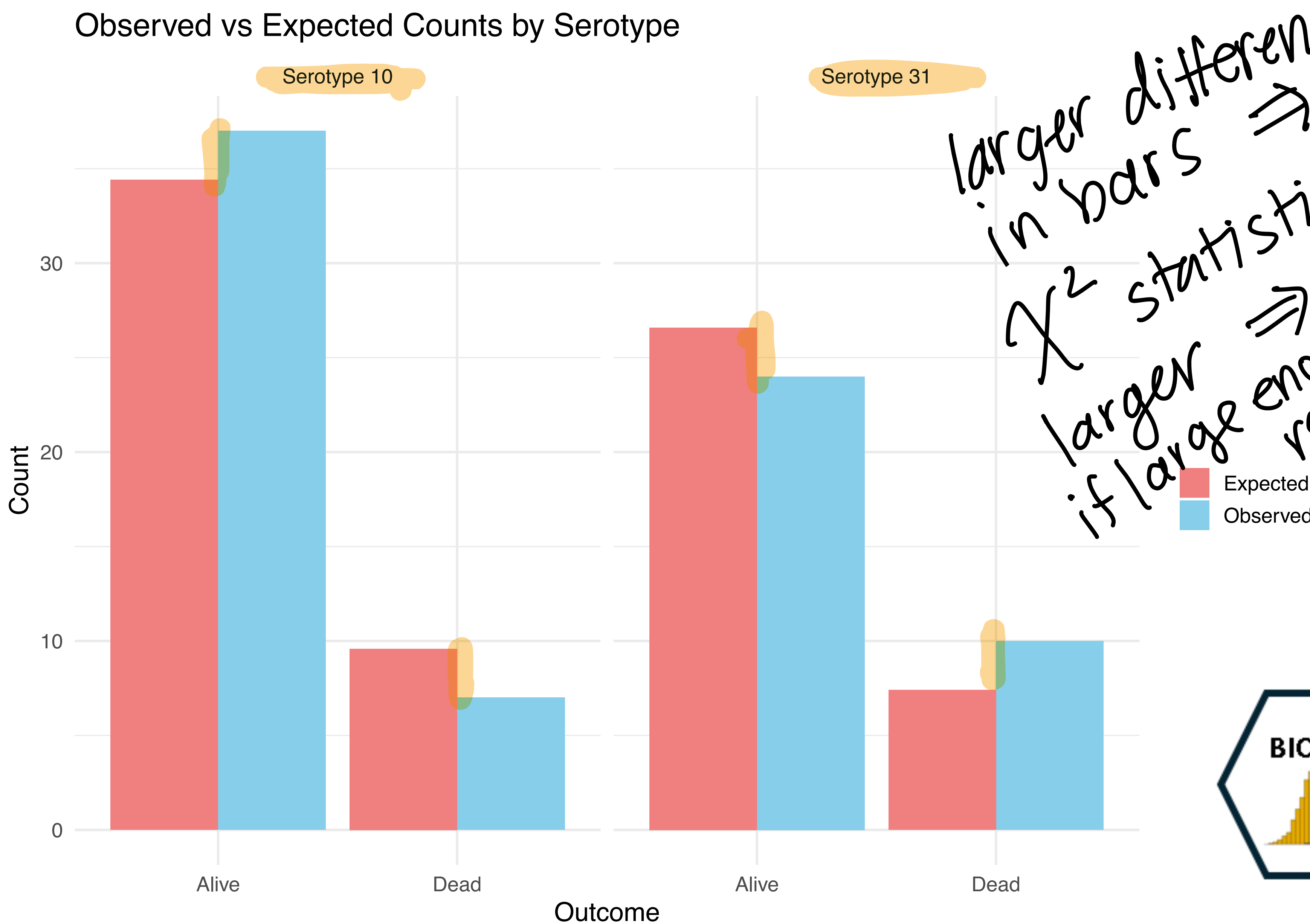
$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

Intuition

- If observed counts are **close** to what we'd expect under independence, χ^2 will be small, then we'd fail to reject H_0.
- If they're **far** from expected, χ^2 will be large, which might lead to evidence of an association.

Visualizing Observed vs Expected Counts

Let's visualize how the **observed** counts compare to the **expected** counts under independence.



larger difference in bars $\Rightarrow$ χ^2 statistic larger $\Rightarrow$ if large enough, reject H_0

Chi-square tests

	Dead	Alive
S 10	1	2
S 31	3	4

$r = 2$

$c = 2$

$r \times c = 4$

The **chi-square test** compares the observed frequencies in each cell to the expected count under H_0; if the total differences are large enough, we reject H_0:

$$\chi^2 = \sum_{i=1}^{r\times c} \frac{(O_i - E_i)^2}{E_i} = \frac{(O_1 - E_1)^2}{E_1} + \frac{(O_2 - E_2)^2}{E_2} + \cdots + \frac{(O_4 - E_4)^2}{E_4}$$

which has a $\chi^2_{(r-1)\times(c-1)}$ distribution, under H_0.

χ^2 has 1 parameter

$(r-1)\cdot(c-1) = (2-1)\cdot(2-1) = 1$

$\boxed{\chi^2_1}$

Chi-square tests

c concatenate

(37, 7, 24, 10)
vector = 1D

```
data <- matrix(c(37, 7, 24, 10),
               byrow = T, nrow = 2)
data
```

, ncol=2)

matrix=2D

↳ two rows

↳ fill in matrix values by ROW.

```
     [,1] [,2]
[1,]   37    7
[2,]   24   10
```

1	2
3	4

1	3
2	4

```
chisq.test(data)
```

```
	Pearson's Chi-squared test with Yates' continuity corre
```

χ^2_1 statistic

= (r-1)·(c-1)

```
data:  data
X-squared = 1.3359, df = 1, p-value = 0.2478
```

What might we conclude at $\alpha = .05$?

Conclusion

- **Conclusion**: Since the p-value is greater than 0.05, we fail to reject the null hypothesis. There is not enough evidence to conclude that survival differs between Serotype 10 and Serotype 31.

Fisher's exact test

- The chi-square test relies on asymptotic results (i.e. large sample size), which might not be appropriate if we have small cell counts (common rules of thumb are >5 or >10 observed in each cell).
- **Fisher's exact test** calculates an exact p-value under a specific distributional assumption.

10	3
15	12

Serotype	Alive	Dead	Total
Serotype 10	37	7	44
Serotype 31	24	10	34
Total	61	17	78

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Fisher's exact test

Fisher's exact test relies on the **hypergeometric distribution** (don't worry about this, seriously).

Serotype	Alive	Dead	Total
Serotype 10	a	c	a + c
Serotype 31	b	d	b + d
Total	a + b	c + d	a + b + c + d

Conditionally on the margins (assumed fixed), then each of the individual cells is distributed according to the hypergeometric distribution. We can thus calculate the exact probabilities of obtaining specific *tables*.

Fisher's exact test

Observed data:

Serotype	Alive	Dead	Total
Serotype 10	37	7	44
Serotype 31	24	10	34
Total	61	17	78

We could use the hypergeometric distribution to calculate the probability of seeing this table, assuming 61 total alive patients, 17 dead, and 44 and 34 infections of serotypes 10 and 31, respectively (lots of factorials, so you could imagine it wasn't fun before modern computers).

Fisher's exact test

Another "possible" table:

Serotype	Alive	Dead	Total
Serotype 10	36	8	44
Serotype 31	25	9	34
Total	61	17	78

- Here we've displayed another table with the **same margins**, but with some slightly different potential cell counts.
- Again we could use the hypergeometric distribution to calculate the probability of observing the cell counts conditionally on fixed margins.

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Fisher's exact test

Fisher's exact test relies on constructing *all possible* contingency tables with the same margins, and then summing up probabilities of all tables as extreme or more extreme than the observed data.

Question

What might "more extreme" mean in the context of contingency tables?

Further away from expected counts under independence.

Fisher's exact test

H_0: The proportion of deaths is the same for Serotype 10 and Serotype 31 (no association between serotype and outcome).

H_A: The proportion of deaths differs between serotypes (there *is* an association)

```
data <- matrix(c(37, 7, 24, 10), byrow = T, nrow = 2)
data
```

```
     [,1] [,2]
[1,]   37    7
[2,]   24   10
```

Conducting the Fisher's Exact test

```r
fisher.test(data)
```

```

	Fisher's Exact Test for Count Data

data:  data
p-value = 0.1758
alternative hypothesis: true odds ratio is not equal to 1
95 percent confidence interval:
 0.6465347 7.7636744
sample estimates:
odds ratio
  2.179552
```

- What might we conclude at $\alpha = 0.05$?

Conclusion

- The Fisher's Exact Test yielded a p-value of 0.17, which is greater than the conventional significance level of 0.05. Therefore, we fail to reject the null hypothesis.
- There is no statistically significant evidence of an association between serotype and survival status in this sample.

Odds ratios

$$\frac{P(\text{death} \mid S_{10})}{P(\text{alive} \mid S_{10})}$$

Serotype	Alive	Dead	Total
Serotype 10	37	7	44
Serotype 31	24	10	34
Total	61	17	78

- Odds of death among serotype 10 patients $= \frac{7/44}{37/44} = 7/37$ 2
- Odds of death among serotype 31 patients $= \frac{10/34}{24/34} = 10/24$ 1
- **Odds ratio** of death comparing 31 to 10 is thus $\frac{10/24}{7/37} \approx 2.2$

$$\frac{\text{Odds death } S_{31}}{\text{Odds death } S_{10}} =$$

3

r by c tables

Take ratio.

Serotype	Alive	Dead	Total
Serotype 10	37	7	44
Serotype 15	60	12	72
Serotype 20	97	9	106
Serotype 31	24	10	34
Total	**218**	**38**	**256**

We can also extend the chi-square test and Fisher's exact test to more general contingency tables (although the notion of an odds ratio no longer makes sense here).

Winter Olympics training

Suppose we were interested in whether a particular training regimen was associated with "success" as measured by whether an athlete qualified for being on the US Winter Olympic Team for giant slalom.

Winter Olympics training

	Qualified	Did Not Qualify
New Regimen	129	97
Old Regimen	112	114

```
    Pearson's Chi-squared test with Yates' continuity corre

data:  data
X-squared = 2.2755, df = 1, p-value = 0.1314
```

Practice

Write out the code to 1) construct the data matrix and 2) to conduct a chi square test in R. What might we conclude using $alpha = .05$?

Overview

Fail to reject H_0

No evidence to suggest association

- Reviewed AE 03 from last time.
- Chi-square test and example
- Fisher exact test and example (use if sample sizes aren't large enough)

Next Class

- Tuesday's class: Non-Parametric Methods
- Enjoy Fall Break!